

157. Read N Characters Given Read4 Premium

Solved ●

Easy Topics 

Given a `file` and assume that you can only read the file using a given method `read4`, implement a method to read `n` characters.

Method `read4`:

The API `read4` reads **four consecutive characters** from `file`, then writes those characters into the buffer array `buf4`.

The return value is the number of actual characters read.

Note that `read4()` has its own file pointer, much like `FILE *fp` in C.

Definition of `read4`:

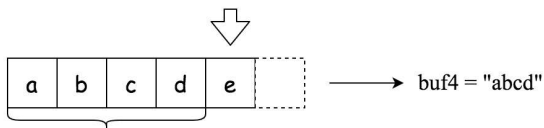
Parameter: `char[] buf4`

Returns: `int`

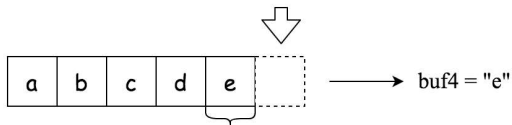
`buf4[]` is a destination, not a source. The results from `read4` will be copied to `buf4[]`.

Below is a high-level example of how `read4` works:

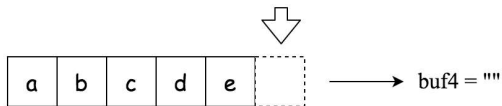
The first call of `read4`



The second call of `read4`



The third / forth / etc calls of `read4`



```
File file("abcde"); // File is "abcde ", initially file pointer (fp) points to 'a'
char[] buf4 = new char[4]; // Create buffer with enough space to store characters
read4(buf4); // read4 returns 4. Now buf4 = "abcd", fp points to 'e'
read4(buf4); // read4 returns 1. Now buf4 = "e", fp points to end of file
read4(buf4); // read4 returns 0. Now buf4 = "", fp points to end of file
```

Method `read`:

By using the `read4` method, implement the method `read` that reads `n` characters from `file` and store it in the buffer array `buf`. Consider that you cannot manipulate `file` directly.

The return value is the number of actual characters read.

Definition of `read`:

Parameters: `char[] buf`, `int n`

Returns: `int`

`buf[]` is a destination, not a source. You will need to write the results to `buf[]`.

Note:

- Consider that you cannot manipulate the file directly. The file is only accessible for `read4` but not for `read`.
- The `read` function will only be called once for each test case.
- You may assume the destination buffer array, `buf`, is guaranteed to have enough space for storing `n` characters.

Example 1:

Input: file = "abc", n = 4

Output: 3

Explanation: After calling your read method, buf should contain "abc". We read a total of 3 characters from the file, so return 3. Note that "abc" is the file's content, not buf. buf is the destination buffer that you will have to write the results to.

Example 2:

Input: file = "abcde", n = 5

Output: 5

Explanation: After calling your read method, buf should contain "abcde". We read a total of 5 characters from the file, so return 5.

Example 3:

Input: file = "abcdABCD1234", n = 12

Output: 12

Explanation: After calling your read method, buf should contain "abcdABCD1234". We read a total of 12 characters from the file, so return 12.

Constraints:

- $1 \leq \text{file.length} \leq 500$
- `file` consist of English letters and digits.
- $1 \leq n \leq 1000$

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Yes No

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Topics



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